



THE AMERICAN
UNIVERSITY IN CAIRO

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CSCE 363/3611 - Digital Signal Processing

Emotion Recognition Using EEG

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Neural Engineering

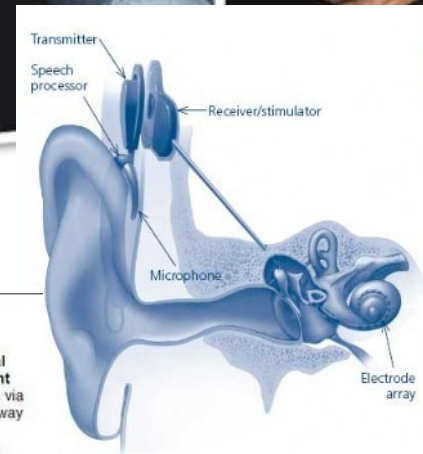
- **Neural Engineering** is a field of research that focuses on engineering methods to investigate the function of the central and peripheral nervous system and manipulate its behavior
- **Neural Interfaces** are systems that can help restore sensory function, communication, and control to impaired humans
- The main principle is that disabled people would have their brains or parts of their brains fully functional
- Neural Interfaces make use of functional parts to restore a lost function
- Objectives of Neural Engineering:
 - 1 Understand Brain Function
 - 2 Provide Therapeutic, Assistive and Augmentative Technology

Neural Engineering

- Examples of Neural Interfaces:



Motor Brain-Machine Interface



Cochlear Implant

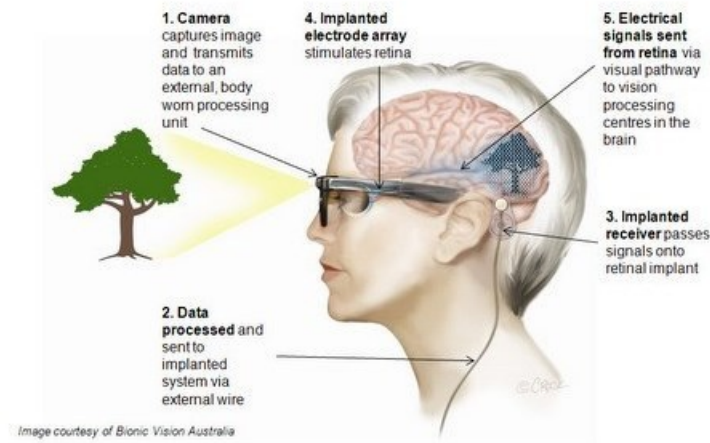
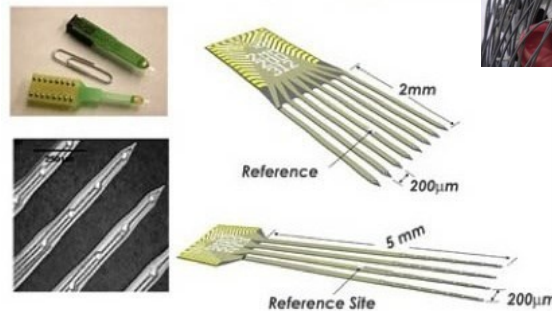
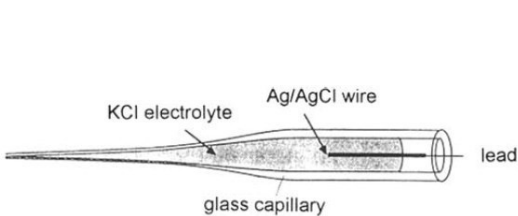
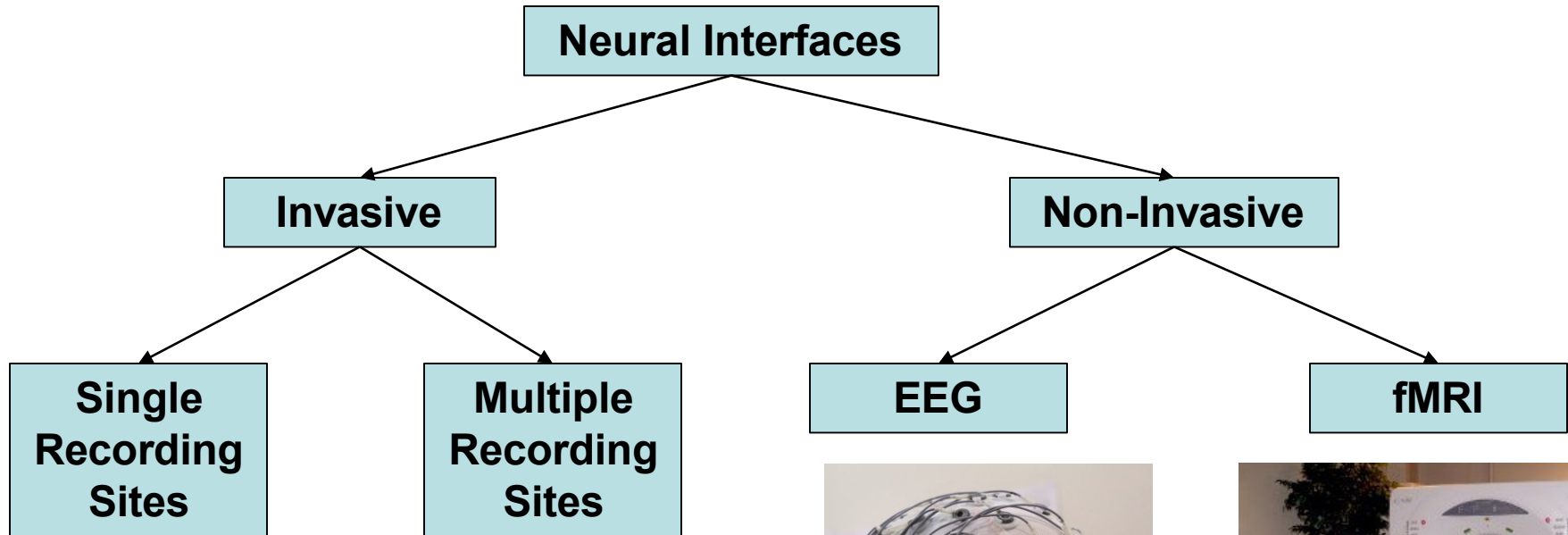


Image courtesy of Bionic Vision Australia

Visual Prosthesis

Neural Engineering

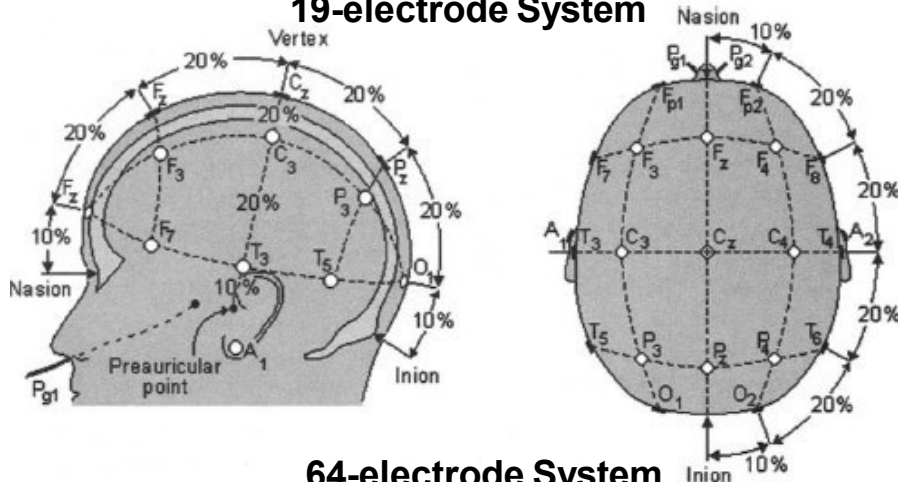
- Types of Neural Interfaces



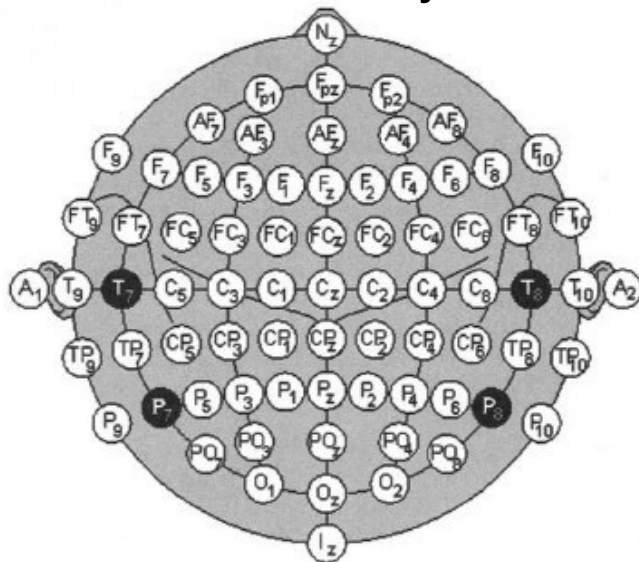
Electroencephalography (EEG)

- Many EEG-based systems use an electrode placement strategy suggested by the **International 10/20 system**

19-electrode System



64-electrode System

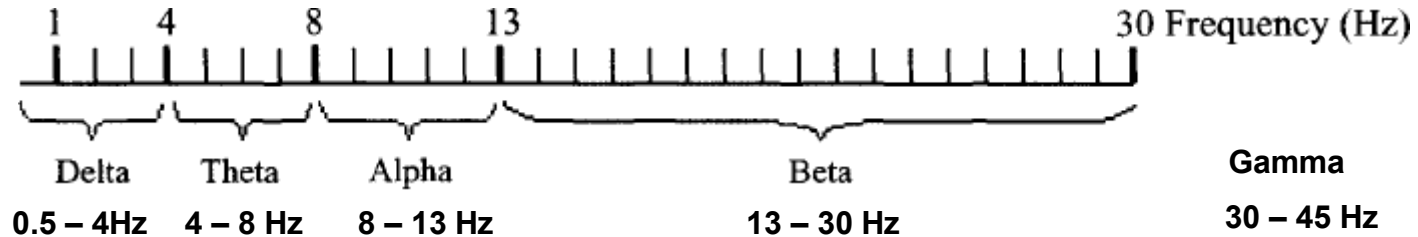


F: Frontal
C: Central
P: Parietal
O: Occipital
T: Temporal

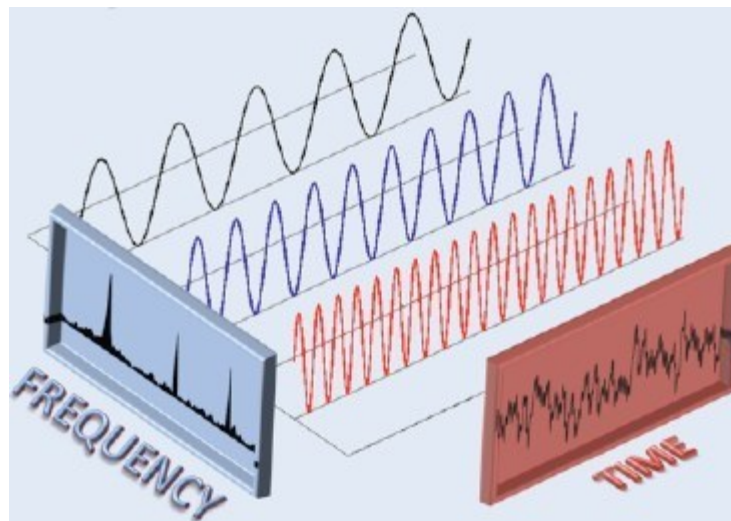


EEG Frequency Bands

- Signals recorded from EEG are split into several bands

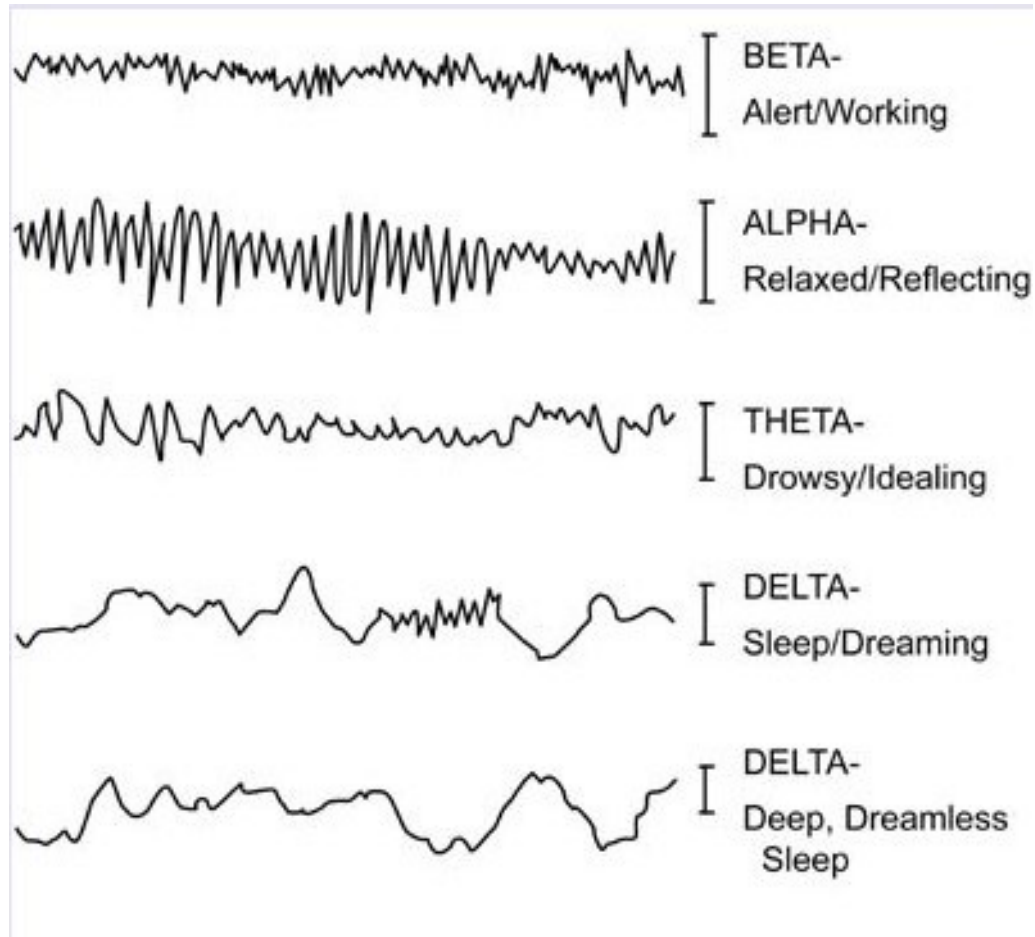


- Each frequency band has some correlation with different mental functions



EEG Frequency Bands

- EEG frequency bands and consciousness



Emotion Arousal-Valence Model

The Arousal-Valence Model of Emotions

Valence (X-axis)

How positive or negative the emotion feels.

High = pleasant (joy, love)

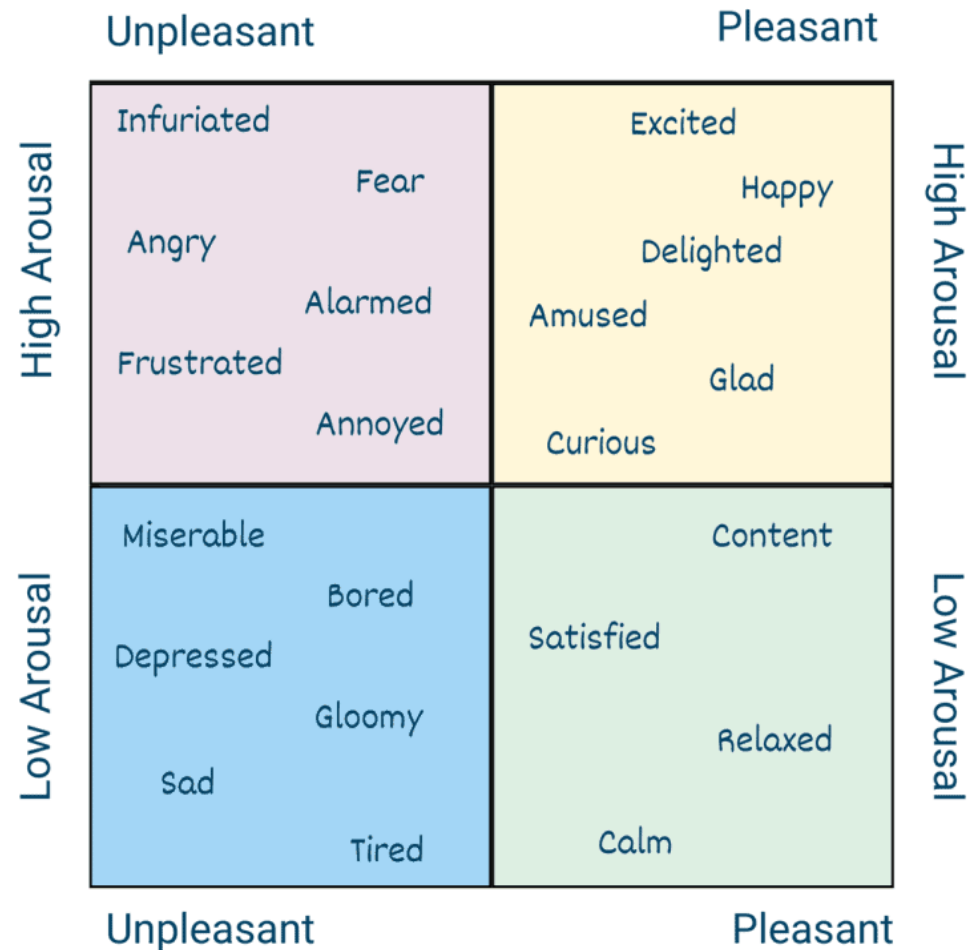
Low = unpleasant (fear, sadness)

Arousal (Y-axis)

The intensity or activation level.

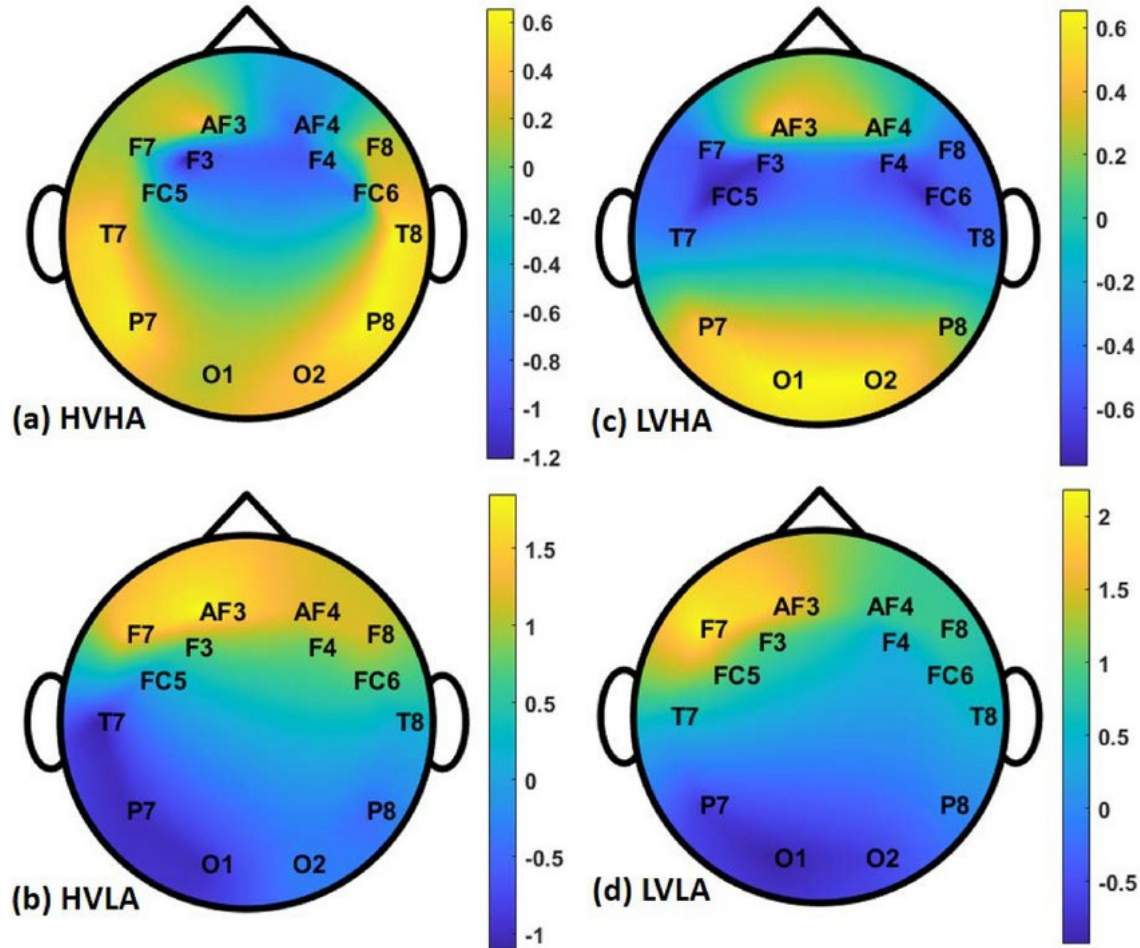
High = excited, alert, energetic

Low = calm, relaxed, sleepy



EEG with Arousal-Valence Model

EEG varies with different emotions



Applications

EEG-Based Emotion Detection While Watching a Video



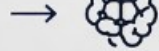
1. Stimulus (Video)

Participant watches a video clip eliciting an emotion.



2. EEG Recording

EEG signals are recorded using an EEG cap while the participant watches.



3. Signal Processing

EEG signals are preprocessed and features are extracted for emotion analysis.



4. Emotion Detection

The model predicts the participant's emotion in real time.

EEG Cap

64-channel EEG cap records brain activity throughout the experiment.

Experiment Goal

To detect the participant's emotional state in real time by analyzing EEG signals while watching an emotion-eliciting video.

EEG Signals (Real Time)

Brain activity is recorded continuously while the participant watches the video.

Detected Emotion (Real Time)

Sadness



Confidence: 0.87

Arousal  0.22
Low High

Valence  0.18
Negative Positive

Algorithm to Implement

Given a dataset of 3 subjects watching 40 different videos, you should do the following:

1. Segment EEG recordings into fixed-length windows of N seconds per video (evaluate multiple N values).
2. Extract features per segment:
 - Time-domain (statistical descriptors)
 - Frequency-domain (band power across standard EEG bands)
3. Construct feature representations:
 - Time-only, Frequency-only, and Combined feature vectors
4. Train and evaluate a K-Nearest Neighbor (KNN) classifier with $K = 1$ to 10 using 5-fold cross-validation
5. Analyze performance of classification for Arousal and Valence:
 - Classification accuracy vs. segment length (N)
 - Classification accuracy vs. number of neighbors (K)

***K*-nearest Neighbor (KNN) Classifier**

- Most basic instance-based method
- Uses Euclidean distance to determine how dissimilar a pair of points are

$$d(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{\sum_{r=1}^n (x_{ir} - x_{jr})^2}$$

- For any new input vector, the nearest K points are considered
- A majority voting scheme is used to classify the new input vector

K-nearest Neighbor (KNN) Classifier

